Slicing Quantum Programs for Comprehension: What a Quantum Slice Should Look Like

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I. Introduction

A developer who sees a qubit end in the wrong state asks which lines of the program determine it. For classical programs a backward slice answers this [1], and programmers use slices to read and debug code [2], [3]. Quantum programs need such help more. Their intermediate states cannot be inspected without disturbing them, and a gate can change a qubit it does not target, as in phase kickback, where a controlled gate changes the phase of its control without entangling the two qubits. Quantum developers debug by printing, drawing circuits, simulating and reading code, and report that this does not scale [4]. Bug detectors such as LintQ and QChecker find known patterns by static analysis, which reads a program without running it, but many faults are wrong outputs that no pattern describes [5], [6].

Slicing has been proposed for quantum programs [7], [8], [9], but no proposed dependence rule has been checked against what a quantum program computes. For comprehension this matters most, because a slice that hides code able to change the qubit sends its reader away from the fault, and such slices look right. The classical slice of the textbook program of Fig. 1 reproduces the faulty program's state and leaves out the faulty line. Over the 784 circuit–qubit pairs of our study, reading a controlled gate as an assignment that leaves its control unchanged gives a wrong slice for 26.0% of the pairs, and 43% of those reproduce the whole program's outcome histogram, so a developer who checks a slice by its outcomes would accept them.

This paper asks what a quantum slice must keep, and which dependence rule produces it. A slice of a set of qubits must preserve their joint state for every input, and an operation is a dependence of them if a change right after it can change that state. One rule meets both, the *write-all rule*, under which a gate writes every qubit it touches, its controls included. On qubits its slice is the backward light cone, whose soundness compilers already rely on [10], [11]. What has been missing is the case for it as a slice, that it keeps every dependence and that nothing else proposed for quantum programs adds one, and a measurement of what each alternative loses. We make the case in proofs and measure the losses in a pre-registered study on 158 benchmark circuits with an exact ground truth, 119 generated circuits and 5,688 seeded-fault units, with real faults from Bugs4Q [12].

The paper makes four contributions. The first is a definition of what a slice of a set of qubits must preserve and of dependence, which separates sound slices from slices that keep every dependence. The second is a proof that the slice of the quantum dependence graph (QDG), defined by the write-all rule and extended along classical bits, is sound for every input, with the induction mechanized in Isabelle/HOL, and keeps every dependence, with evidence that no standard two-qubit gate can safely write fewer qubits. The third shows that classical data, classical control and quantum data dependence suffice, since apparent quantum control dependence is quantum data dependence and entanglement is neither necessary nor sufficient for dependence. The fourth measures the alternatives. The classical PDG fails almost only through its reading of the control, the QPDG of Yu and Zhao [7] keeps 2.7 times as much code that cannot matter as the QDG, whose slices come within 4.8 points of the smallest slice that keeps every dependence, and pruning drops dependences. Slices narrow where a developer looks but improve neither mutation-based fault ranking nor a language model's localization of faults.

The paper builds no slicer for a particular language. Its definitions are stated on the circuit a program builds, Section IV-E shows how a slice reaches source lines, and computing slices at scale on the source developers write is a separate question. Section II reviews related work, Section III works through Fig. 1, Section IV defines the semantics and the rules, Sections V and VI present the study, and the last three sections give the implications, the threats to validity and the conclusion. The data, scripts, protocol and proofs are in an anonymous replication package.[TODO: anonymous package link]

II. Related Work

This section reviews four lines of research and closes with the gaps they leave open and with Table I, which compares the dependence models proposed or used for quantum programs.

A. Program slicing and its semantics

Weiser introduced program slicing [1] and showed that programmers use slices when they debug [2], and slicing is a comprehension aid as much as a debugging one [3]. Dependence graphs reduced slicing to reachability [13], [14], [15], Podgurski and Clarke defined dependence semantically [16], and dynamic slicing restricts a slice to one execution [17], [18]. A backward slice gives the statements that can affect a value at a point, and its dual, the forward slice, the statements a given statement can affect [19], [15]. Slicers need not build a graph, as srcSlice computes dependences from a slice profile of every variable [20], and surveys catalog the kinds of slices that followed [19], [21], [22]. Probabilistic slicing preserves the distribution of the criterion variables [23], [24], [25],

and Gallagher names quantum programs as a direction for semantically defined slicing [26].

B. Dependence models for quantum programs

Yu and Zhao's quantum program dependence graph (QPDG) [7] extends the classical PDG [14] with entanglement, no-cloning, temporal and measurement dependences and proposes slicing on it. It is defined in prose with examples and has been neither implemented nor evaluated, and in the graph drawn for its first listing the slice of a measured qubit omits the two gates that fix its outcome, so the slice returns 1 with probability 1/2 where the program returns 1 with certainty. Kaul et al.'s quantum code property graph (QCPG) for Qiskit and OpenQASM is queried for bugs and smells, among them gates that do not affect a measured qubit [8]. Its data-flow edges run from a CNOT's control to its target only, since the authors state that the control is unchanged, so the query can report gates whose effect reaches a measured qubit through a control. The graph is demonstrated on example programs. Behler et al. derive the order of qubit operations from OpenQASM 3 source to detect patterns and suggest refactorings, and their order table marks a CNOT's control apart from its target [27], [28]. Murillo et al. describe the QPDG as an aid to understanding programs [9], and Ramalho et al. find dependence analysis of circuits almost unexplored [29]. Entanglement analyses approximate which qubits may be correlated, with type systems [30], [31], [32], abstract interpretation [33], [34], the stabilizer formalism [35] and data-flow rules [36], and ScaffCC assumes that interacting qubits may be entangled [37]. None of them slices.

C. Light cones and compilers

The *backward light cone* of a set of qubits at the end of a circuit is the set of gates that can affect their final state. Compilers schedule and map circuits on gate-dependence graphs built on it [10], [38], QAOA analyses use it to bound the gates that can reach an observable [39], and quantum separation logics reason locally about the qubits a program touches [40], [41]. Qiskit's `LightCone` pass [11] walks a circuit backward from given qubits or a Pauli observable, keeps the gates in the light cone and also removes each gate that commutes with the observable, since such a gate cannot change its expectation value. It decides commutation for the circuit's actual angles and rejects circuits with mid-circuit measurements. Heim et al. likewise argue that quantum dependences should account for commutation [42]. Compilers also call a circuit's time windows its slices [43], a sense different from ours.

D. Debugging tools and bug detectors

Metwalli and Van Meter split a circuit at barriers and remove the qubits a part does not use [44], [45], and Qiskit's circuit-slicing utilities cut a circuit by barriers, depth or gate type [46]. Neither follows dependences. Bug detectors find known patterns [47], [6], [48]. LintQ orders two operations that share a qubit [5], which is the relation of the write-all rule, but uses it for patterns, not slices. Of the 42 Bugs4Q bugs on which QChecker was evaluated, 22 are wrong outputs that no pattern detects [6], and a slice shows where such a fault can be. We reuse Bugs4Q [12], mutation tools [49], [50], [51] and mutation-based fault localization [52].

E. Open gaps and the data used

Table I compares the dependence models these works use. Three gaps remain. First, no rule proposed for quantum slicing has been checked against what a quantum program computes, and the only one worked out in detail, the QPDG's, gives a wrong slice. Second, the QCPG and the order tables of Behler et al. read a gate's control as unchanged, and no work has measured what that does to a slice. Third, the light cone is known to be sound, but no work has asked whether it keeps every dependence, how far it is from the smallest slice that does, and what smaller rules lose. We keep the classical notions of a criterion, a backward and a forward slice and dependence as the possibility of change, and define them for the joint state of a set of qubits, which unlike a distribution in one basis keeps phase. Section IV-C shows that entanglement neither implies nor is implied by dependence, so entanglement analyses cannot decide what a slice keeps.

The dependence graphs were shown on examples and the detectors evaluated on Bugs4Q and on LintQ's GitHub corpus, but no rule for quantum slicing has been evaluated against a ground truth. We use Bugs4Q for real faults and add circuits with a simulated ground truth and seeded faults.

III. A Classical Slice That Looks Right

This section shows, on a textbook program, how a slice computed the classical way can reproduce the output a developer sees and still hide the fault, and why a check of measured outcomes cannot tell. It starts with the quantum concepts the rest of the paper uses.

Background: A qubit's state is a superposition $\alpha|0\rangle + \beta|1\rangle$, and measuring it yields 0 or 1 with probabilities $|\alpha|^2$ and $|\beta|^2$. The states $|+\rangle$ and $|-\rangle$ give both outcomes equally often and differ only in phase, which no measurement in the computational basis can see, and the Hadamard gate H turns $|0\rangle$ into $|+\rangle$ and back. The controlled-NOT (CNOT) flips its target when its control is 1, but when the target is in $|-\rangle$ it leaves the target unchanged and flips the phase of its control instead. This is phase kickback, and it needs no entanglement, the state of two qubits that cannot be written as a state of each alone. A measurement writes a classical bit that later gates may read. Qiskit programs build circuits in Python, with loops and functions that compute which gates act on which qubits.

The program: The Bernstein–Vazirani algorithm [54] finds a hidden bit string s with one query to an oracle. Fig. 1 shows the program of the Qiskit textbook [53], with $s = 011$. Lines 24 and 25 prepare the auxiliary qubit q_3 in $|-\rangle$, line 29 puts every data qubit in $|+\rangle$, and for every bit of s that is 1, the CNOT of line 40 kicks a phase onto its data qubit, turning $|+\rangle$ into $|-\rangle$. The Hadamard gates of line 47 turn these phases into

TABLE I
DEPENDENCE MODELS FOR QUANTUM PROGRAMS, BY WHAT THEY READ A GATE TO DO AND WHAT THEY GUARANTEE

Approach	CNOT writes	Measured bits	Extra edges	Slice semantics	Guarantee	Evaluated on
QPDG, Yu and Zhao [7]	?	●	entanglement, closed	○	none	5 listings, not implemented
QCPG, Kaul et al. [8]	t	●	none	○	none	3 listings, 1 Qiskit/OpenQASM circuit
Behler et al. [27], [28]	t	◐	none	○	none	16 examples of OpenQASM 3
LintQ [5]	c, t	●	none	○	none	7,568 GitHub programs, 216 confirmed bugs
TriQ, staq [10], [38]	c, t	–	none	○	sound, known	12 and 24 benchmark circuits
Qiskit `LightCone` [11]	c, t	○	none, removes commuting gates	◐	observable, fixed angles	–
QDG, this paper	c, t	●	none	●	sound, complete	354 circuits of MQT Bench, QASMBench, VeriQBench and our own, 5,703 fault units incl. 15 from Bugs4Q

● has the property, ◐ has it in part, ○ lacks it, ? means the authors do not specify it, and – means it does not apply. *CNOT writes*, the qubits a CNOT with control c and target t is read to change, from which the dependences follow. *Measured bits*, classical control is followed from a measurement to the operations its bit conditions. *Extra edges*, dependences added beyond those of the gates' qubits. *Slice semantics*, what a slice must preserve is defined. *Guarantee*, the property of a slice that the authors state or prove; complete means that it keeps every dependence. *Evaluated on*, the data of the original evaluation.

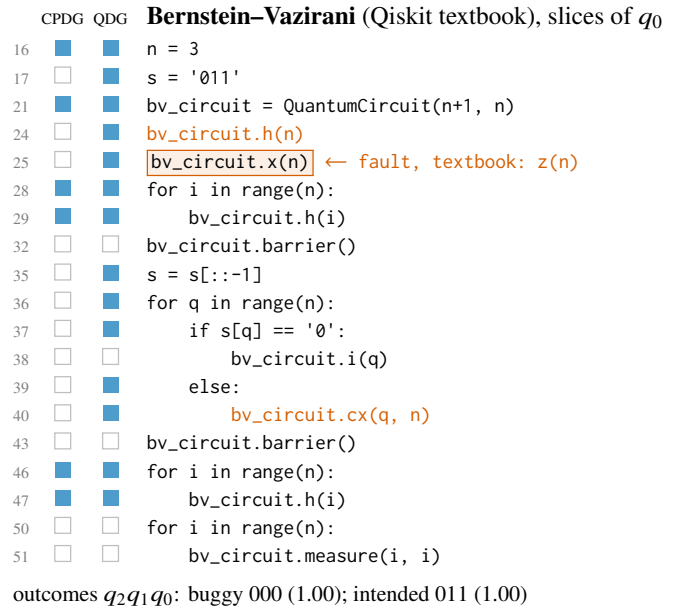


Fig. 1. The Bernstein–Vazirani program of the Qiskit textbook [53], with comments and blank lines omitted and the textbook's line numbers kept. Line 25 carries a seeded fault, `x(n)` where the textbook applies `z(n)`. The two columns mark the lines in the backward slice of q_0 under the classical PDG and under the QDG. The classical slice misses the orange lines, which can change q_0. Each slice holds the lines of its operations and the host lines that compute their qubits, as Section IV-E describes. Slices, outcomes and dependences are computed by the study's code.

the bits of s. We seeded one fault, X instead of Z on line 25, which leaves the auxiliary in $|+\rangle$, on which a CNOT changes nothing, so the program returns 000 instead of 011.

The slices: The developer sees q_0 read 0 where s says 1 and slices backward from q_0. A classical slicer reads `cx(q, n)` as the assignment $q_n \leftarrow q_n \oplus q$, which reads q and writes only the auxiliary. Its slice of q_0, the left column of Fig. 1, keeps the Hadamard gates of lines 29 and 47 and the lines that compute their qubits, 16, 21, 28 and 46. The slice looks right. Its 2 operations give q_0 the same state as the faulty program, with fidelity 1.00, a similarity of quantum states that is 1 exactly when they are equal, so nothing sends the developer to lines 24, 25 and 40, where the fault is.

Yet these lines can change q_0. Restoring Z on line 25 turns q_0 from $|0\rangle$ into $|1\rangle$, states with fidelity 0.00, because the CNOT acts on its control as well as its target. The QDG of Section IV, which lets every gate write all its qubits, keeps lines 24, 25, 29, 40 and 47 and the lines that compute them, among them the oracle's loop and condition on lines 35 to 39, and so contains the fault. The QPDG of Yu and Zhao keeps the same lines here, since its two extra operations are later iterations of lines 29 and 40, and Fig. 2 shows where it costs more.

Why the outcomes cannot tell: Here the fault shows in the outcome of q_0, but not always. In a real bug from Bugs4Q [12], posted on Stack Overflow, a program meant to prepare $(|01\rangle+|10\rangle)/\sqrt{2}$ applies `h(0)`, `h(1)` and `cx(0,1)`, with `h(1)` where `x(1)` was needed, and returns all four bit strings equally often. Each qubit alone reads 1 with probability 0.5 in the buggy and the intended program alike, so only joint outcomes reveal the failure. The fault does change the state of q_0, a uniformly random bit in the intended program and the pure $|+\rangle$ in the buggy one, with fidelity 0.50. Because the CNOT writes its control, the QDG slice of q_0 contains the faulty `h(1)`, while the classical slice keeps only `h(0)`.

Section VI shows that both lessons generalize. Reading a gate as an assignment misses influence that flows into a control, and judging a slice by one qubit's outcomes misses errors in phase and correlation, so a slice must be judged by the state of its qubits, as Section IV defines it.

IV. QUANTUM DEPENDENCE AND THE WRITE-ALL RULE

This section defines what a slice must preserve and what a dependence is, states the write-all rule and its guarantees,

shows that three kinds of dependence suffice, and places every other rule relative to it.

A. Circuits, criteria and dependence

A Qiskit program builds a *circuit*, which a simulator or a quantum computer then runs. The circuit $P = o_0 \dots o_{L-1}$ is the sequence of operations the program appends, in order, on n qubits $Q = \{0, \dots, n-1\}$ and a set C of classical bits that store measurement outcomes.

The qubits and bits an operation touches are its *locations* $loc(o) \subseteq Q \cup C$. A gate touches the qubits it is applied to, so `h(0)` has the location q_0 and `cx(0, 1)` the locations q_0 and q_1. A measurement touches the measured qubit and the bit that receives its outcome, so `measure(2, 0)` has the locations q_2 and c_0. A reset, which returns a qubit to $|0\rangle$, touches that qubit. A conditioned gate is applied only when a measured bit has a given value, as in `x(0)` inside `if_test((c[0], 1))`, and it touches its qubits and the bits its condition reads, here q_0 and c_0.

The program's own loops and branches run while the circuit is built and decide which operations it contains. Loops that run on the device, inside the circuit, are outside this model, and none of the study's circuits has one.

The state of the qubits is a density matrix ρ, which also covers the mixtures that measurements create. Running P on the input ρ, with every bit initially 0, gives $[\![P]\!](\rho)$, the final joint state of qubits and bits, averaged over the outcomes of the measurements. A developer asks about some of the qubits, a set $W \subseteq Q$. Their *reduced state* $\rho_W = \mathrm{Tr}_{\overline{W}}(\rho)$ is what remains of ρ when everything else, $\overline{W} = (Q \setminus W) \cup C$, is traced out, that is, averaged away. It determines every statistic of measuring W in any basis. A slice is a set S of positions in P, judged by running the restriction $P|_S$, which keeps only the operations at those positions, in order.

Definition 1 (Slicing criterion): A slicing criterion is a nonempty set $W \subseteq Q$ of qubits, whose state is taken at the end of the circuit, before any terminal measurement reads it.

The criterion is a set because the joint state of entangled qubits is not determined by their separate states. Yu and Zhao keep the classical criterion of a variable at a statement [7], and we preserve the joint state instead, taken before readout so that a slice stays correct when composed with later gates or read in another basis.

Definition 2 (Soundness): A set of positions S *preserves* W at input ρ if $\mathrm{Tr}_{\overline{W}}[\![P|_S]\!](\rho) = \mathrm{Tr}_{\overline{W}}[\![P]\!](\rho)$. It is a *sound* slice for W if it preserves W at every input.

Comprehension needs more than soundness, the assurance that a slice shows every place where a change, and so a fault, could alter the criterion.

Definition 3 (Dependence): Operation o_k is a *dependence* of W in P if there are an input ρ and a unitary U on the qubits of o_k such that inserting U immediately after o_k changes $\mathrm{Tr}_{\overline{W}}[\![P]\!](\rho)$. For a measurement, negating the bit it writes also counts as such a change. $Dep_W(P)$ is the set of dependences of W, and a set S is *dependence-complete* for W if $Dep_W(P) \subseteq S$.

This is semantic dependence [16] stated for quantum states. A sound slice need not contain every dependence. The identity gate of line 38 in Fig. 1, which acts on q_2, is a dependence of q_2, since a perturbation inserted after it changes the state of q_2, yet deleting it changes nothing. Conversely, deleting a single unitary that is not a dependence never changes ρ_W, since it has the same effect as inserting its inverse.

B. The write-all rule

A *dependence rule* R assigns each operation the locations it reads, $r_R(o)$, and the locations it writes, $w_R(o)$. Operation o_j depends on o_i under R if $i < j$ and some location in $w_R(o_i) \cap r_R(o_j)$ is not written by an operation between them. The backward slice $B_R(W)$ is the set of operations from which the last writer of some $q \in W$ is reachable along these dependences. The rules differ in one choice, what an operation writes.

Definition 4 (Write-all rule): Every operation reads and writes all of its locations, $r(o) = w(o) = loc(o)$. In particular a controlled gate writes its controls as well as its targets. The *quantum dependence graph* (QDG) has the operations as nodes and the dependences of this rule as edges, and the QDG slice $B_{\mathrm{QDG}}(W)$ is backward reachability from W.

The QDG specifies the slice, and computing it needs no graph, only one backward pass. Start at the end of the circuit with the needed set $N = W$, and whenever an operation acts on a location in N, keep it and add all its locations to N. On qubits this is the backward light cone of W, which the classical bits extend from a measurement to the operations its bit conditions. Take the program of Fig. 1 with a one-bit secret, the circuit

$$P_1 \;=\; \mathrm{H}(q_1)\;\; \mathrm{Z}(q_1)\;\; \mathrm{H}(q_0)\;\; \mathrm{CX}(q_0, q_1)\;\; \mathrm{H}(q_0)$$

whose gates are those of lines 24, 25, 29, 40 and 47, and the criterion $\{q_0\}$. Walking backward, the CNOT acts on q_0, so q_1 joins N and the slice is the whole circuit, as it must be, since through kickback the gates on q_1 turn q_0 into $|1\rangle$. A rule that lets the CNOT write only q_1 keeps just the Hadamard gates of lines 29 and 47.

On qubits, soundness is a known property of the light cone [10], [39]. We prove it for the QDG, classical bits included, because the results that follow build on the proof.

Theorem 1 (The QDG is sound): For every circuit P and criterion W, $B_{\mathrm{QDG}}(W)$ is a sound slice for W.

Proof sketch: Process P backward with the needed set N, initially W. Operation o_k joins the slice, and N grows by $loc(o_k)$, exactly when $loc(o_k) \cap N \neq \emptyset$, which computes $B_{\mathrm{QDG}}(W)$. Every gate, reset, measurement and conditioned gate is a channel that acts only on $loc(o)$. By locality of the partial trace, the no-signaling principle, an operation outside the slice cannot change the reduced state of the needed locations after it, and an operation in the slice maps the reduced state of the needed set before it to the one after it. Induction over P gives the claim. The induction is mechanized in Isabelle/HOL for an abstract model of local operations, and

that quantum operations satisfy the model is argued on paper. ∎

Proposition 1 (The QDG is dependence-complete): $Dep_W(P) \subseteq B_{\mathrm{QDG}}(W)$.

Proof: Let $o_k \notin B_{\mathrm{QDG}}(W)$, and let P' insert a unitary U after o_k. In the backward pass, no location of o_k is needed after it, since otherwise o_k would be kept. U acts only on locations of o_k, so U is not kept either and the needed set is unchanged. Hence P' and P have the same QDG slice, and by Theorem 1 the same reduced state of W. A negated outcome is handled alike, as an operation on the bit of a measurement outside the slice. ∎

Proposition 2 (The QDG distributes over criteria): $B_{\mathrm{QDG}}(W_1 \cup W_2) = B_{\mathrm{QDG}}(W_1) \cup B_{\mathrm{QDG}}(W_2)$.

Proof: In the backward pass of Theorem 1, the needed set for $W_1 \cup W_2$ is at every position the union of those for W_1 and W_2. ∎

So the QDG slice reproduces the criterion for every input and keeps every dependence, and a tool can combine single-qubit slices. It is not always the smallest such set, since an operation can touch a needed qubit without changing it, and Section VI measures how close it comes to the dependences.

No gate can safely write fewer qubits. For each of the 21 two-qubit gates of Qiskit's standard library, a change of either qubit's input changes the other's reduced state for some product input, in an exploratory check of Section V.

C. Three kinds of dependence suffice

The write-all rule needs only three kinds of dependence, all of which the variational ansatz of Fig. 2a, described in Section IV-D, shows. *Classical data* dependence follows the host values that compute gate arguments, such as angles, sizes and qubit indices, there `n`, `layers` and the angles. *Classical control* dependence follows what decides whether an operation is applied at all. It covers the host program's loops and branches, which shape the circuit as it is built, such as `for qubit in range(n)`, and the measured bits that condition operations when the circuit runs, such as the flip conditioned on the measured bit. *Quantum data* dependence follows a qubit through the operations that act on it, under the write-all rule, so that `cx(qubit, qubit + 1)` writes both of its qubits.

A controlled gate looks like a fourth kind, quantum control dependence, since it reads like a branch. It is not one. Its control is coherent, so the gate acts on a superposition of control values, no branch is taken, and it can change its control, as P_1 shows. What looks like control dependence is quantum data dependence in both directions, and reading the gate as a branch is the error of the classical PDG. Within the circuit, control dependence arises only through measurement.

Nor do the other proposed kinds add a dependence. Entanglement is the most natural candidate, but it decides nothing.

Proposition 3 (Entanglement is neither necessary nor sufficient for dependence): (i) An operation can be a dependence of q although, at an input that witnesses the dependence, q is never entangled with its qubits. (ii) An operation can act on a qubit entangled with q and not be a dependence of q.

Proof: (i) In P_1 at input $|00\rangle$, the state is a product state after every operation, yet the gates on q_1 are dependences of q_0, since inserting $\mathrm{Z}(q_1)$ after them changes the final state of q_0. (ii) In $\mathrm{H}(q_0)\,\mathrm{CX}(q_0, q_1)\,\mathrm{X}(q_1)$ the two qubits are maximally entangled before $\mathrm{X}(q_1)$, but by no-signaling no operation on q_1 after the CNOT can change ρ_{q_0}. ∎

By Proposition 1, entanglement edges add no dependence the QDG misses, and by (ii) they can add operations that are not dependences. A CNOT that copies a basis state, the case no-cloning concerns, has the edges of quantum data dependence already. Timing decides when an operation runs, not what a state depends on [43], and superposition changes amplitudes, not the locations an operation acts on [55].

TABLE II
THE DEPENDENCE RULES, BY WHAT A CNOT WITH CONTROL c AND TARGET t WRITES AND BY HOW EACH DIFFERS FROM THE QDG

Rule	$w(cx_{c,t})$	**Relation to the QDG**	**Guarantee**	**Ops**	F
QDG	$\{c, t\}$	**write-all rule**	**sound, complete**	16	1.00
Def-use	$\{t\}$	− control write, − bits	none	5	0.55
Classical PDG	$\{t\}$	− control write	none	7	0.97
QPDG [7]	$\{c, t\}$	+ closed entanglement edges	sound	19	1.00
QPDG, read-only	$\{t\}$	− control write, + entanglement	none	19	1.00

Every rule reads $\{c, t\}$, and complete means dependence-complete. Commutation and state-based pruning remove operations from the QDG slice. Ops is the number of the 19 operations of Fig. 2 that the slice of q_0 keeps, and F its fidelity at $|0 \ldots 0\rangle$.

D. The other rules, relative to the write-all rule

Table II gives each alternative rule by how it differs from the QDG, and Fig. 2 applies them to a program that needs every kind of dependence, an ansatz, the parameterized circuit of a variational algorithm. It has layers of rotations and a chain of CNOTs, after which an auxiliary qubit is measured mid-circuit and q_0 is flipped when the outcome is 1.

The classical PDG and def-use: The classical PDG (CPDG) applies the program dependence graph [14] to the circuit, reading a controlled gate as an assignment that writes only its targets, the usual explanation of a CNOT [29], which holds for computational basis states. A measurement writes its bit, and a conditioned gate reads the bits its condition tests. *Def-use* (DU) is the same without the bits.

Proposition 4 (The classical PDG is unsound): There are circuits for which neither $B_{\mathrm{CPDG}}(W)$ nor $B_{\mathrm{DU}}(W)$ preserves W, even at $|0 \ldots 0\rangle$.

The ansatz is a witness for both. Def-use ignores the bits, so its slice of q_0 keeps only 5 of the 19 operations, with fidelity 0.55, and loses the auxiliary's gate and measurement, which the classical PDG reaches through the green control edge of

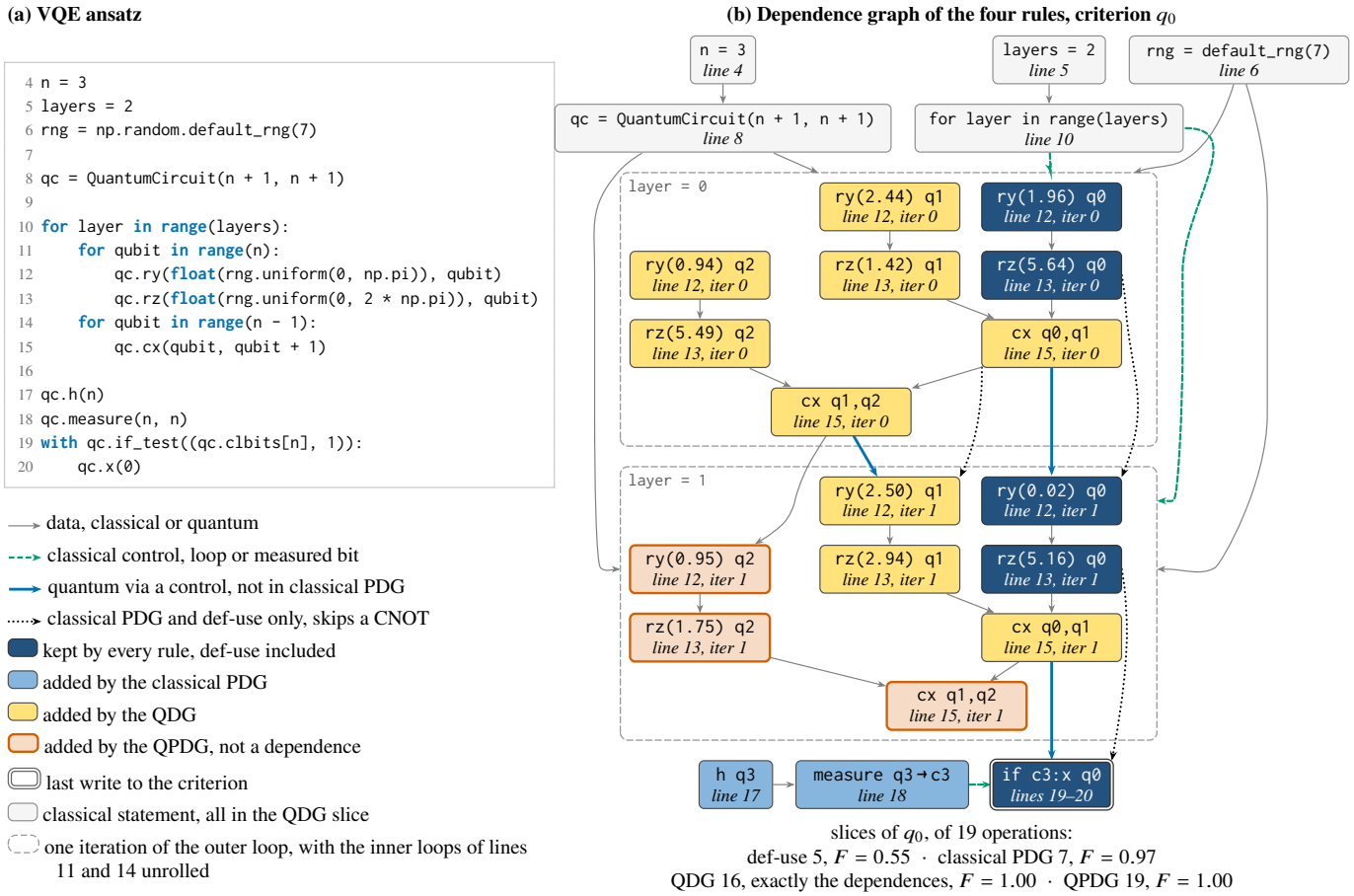


Fig. 2. The rules on a variational ansatz, criterion q_0. (a) The program, described in Section IV-D. (b) One graph for the first four rules of Table II. Each node is an operation as it executes, with its source line and loop iteration, and a box holds each iteration of the outer loop. Grey data edges, classical or quantum, belong to every rule. Green dashed edges are classical control, from the loop of line 10 and from the measurement to the conditioned flip, and def-use lacks the second. Blue edges are quantum data dependence through a CNOT's control. The classical PDG and def-use lack them and have the black dotted edges instead, each of which skips a CNOT whose control they read as unchanged. A node is shaded by the weakest rule that keeps it. The QDG keeps exactly the dependences of q_0, and the QPDG adds the vermilion operations, which are not dependences. Graphs, slices and fidelities at $|0\ldots0\rangle$ are computed by the study's code from the program shown.

Fig. 2b. Each `cx(qubit, qubit + 1)` also changes its control, so q_0 depends on the rotations of q_1 and, from the second layer on, of q_2, which the classical PDG skips with its dotted edges. It keeps 7 operations, misses 9 dependences and reaches fidelity 0.97, while the QDG keeps the 16 dependences of q_0 and no other operation.

The QPDG: Our reimplementation of Yu and Zhao's graph [7] follows readings of its prose logged before any run. Data and control dependence are as in the QDG. For entanglement, which the QPDG does not define, qubits joined by a two-qubit gate anywhere in the circuit are entangled [37], and an edge links every operation to every later one on qubits of the same component. A flow-sensitive variant counts qubits as entangled only from their first coupling gate. Either way the QPDG is the QDG plus entanglement edges, which in Fig. 2b add the three vermilion operations of q_2 that run after the last gate carrying information from q_2 to q_0. Yu and Zhao do not say whether a gate writes its control, so an exploratory variant reads it as unchanged.

Pruning: Two refinements remove operations from the QDG slice. *Commutation pruning*, which Qiskit's `LightCone` pass implements [11], removes each operation that commutes with the Pauli observables X, Y and Z of every criterion qubit propagated backward. *State-based pruning* fixes one input, by default $|0\ldots0\rangle$, and removes an operation that does not act on the criterion if the mutual information between the criterion and the operation's qubits is below 10^{-9} and the removal still preserves W at that input. Neither is dependence-complete, and a state-based slice holds for its one input only, like a conditioned slice [56].

E. Forward slices and source lines

The forward slice $F_{\mathrm{QDG}}(o_k)$ of an operation, the operations reachable from it in the QDG, answers which later operations and qubits a change to o_k can reach. Both slices follow the same paths, so o_k reaches the last operation on a qubit q exactly when $o_k \in B_{\mathrm{QDG}}(\{q\})$, and by Proposition 1 it reaches every qubit of which it is a dependence.

A slice is computed on the circuit and read on the program. Every operation records the source line and loop iterations that appended it, so a slice of operations maps to lines, and the host code that computes the kept operations' qubits, angles and loop bounds follows by a classical slice of the host program [1]. A line is in a slice if one of its operations is, so a single iteration can put it there. The second-layer rotations of q_2 in the ansatz are not dependences of q_0, yet their lines are in its slice through the first layer, so a line slice should name the iterations that put each line in it. The record can be taken while the program builds its circuit, as the study does for Bugs4Q, or by a static analysis of the source, as Behler et al. derive operation order from OpenQASM 3 [28]. Where a value is unknown, such an analysis must assume every qubit an expression may denote and every branch a condition may take, which can only enlarge the slice.

V. Study Design

The study measures every rule of Section IV against its definitions, asking four questions a developer has of a slice.

- **RQ1** Which dependence rules give sound and dependence-complete slices?
- **RQ2** How much smaller than the QDG slice can a slice safely be, and under which inputs?
- **RQ3** What does each rule do to fault localization?
- **RQ4** What does each rule cost in time as circuits grow?

A. Pre-registration

The protocol was committed before the runs reported here. It extends an earlier run whose results it reuses, so the outcomes of H1 and H2 were known when it was written, and every later change is a dated deviation. The five confirmatory tests concern state-based pruning. Every other result is exploratory or descriptive, including the classical PDG's share of wrong slices and everything added after the first results, the classical PDG and `LightCone` baselines, the dependence set and its nesting with the rules, the symbolic-angle, flow-sensitive and read-only variants, and the gate check. Two descriptive tasks for language models were pre-registered, and the most expensive model was dropped for cost before any scored call.

B. Subjects

The rules are compared on circuits, where exact simulation gives the ground truth. Corpus A collects 158 benchmark circuits from MQT Bench [57], QASMBench [58] and VeriQBench [59], with 1 to 14 qubits and a median of 14 operations, at most 126. Every qubit of every circuit is a criterion, which gives 784 circuit–qubit pairs, 80 of them in dynamic circuits with mid-circuit measurement, reset or conditioned gates. The 119 generated circuits come in families of known structure that grow in depth and in width, with 2 to 10 qubits and up to 2,119 operations, and 22 width sweeps extend two of the families to 14 qubits. Corpus L has 55 wider circuits from the same suites, with 15 to 20 qubits and up to 783 operations, on which only some rules finish. Faults are seeded into corpus A, and real faults come from Bugs4Q [12]. A unit is a fault and a qubit whose measured distribution it changes.

C. Ground truth

An independent oracle simulates each circuit exactly, branching over measurement outcomes, and agrees with Qiskit's simulation to within 2.2×10^{-16}. A slice preserves the criterion when the squared Uhlmann fidelity of the criterion's reduced states under the slice and the whole circuit is at least $1-10^{-6}$. The dependences of Definition 3 are measured by inserting two random unitaries after every operation and comparing the criterion's state at 11 inputs, $|0\dots0\rangle$ and random product states. Being sampled, the measured dependence set can only miss true dependences. A catalog of 24 hand-derived cases checks the ground truth and every slicer before the runs.

D. Slicers

Each rule is implemented on circuits from its read and write sets or its pruning step. Qiskit's `LightCone` pass applies to 732 of the 784 pairs. It rejects mid-circuit measurements on 42 pairs, and on 10 its output cannot be matched back to the original. Exploratory variants make commutation pruning hold for all angles and give state-based pruning a joint criterion.

E. Robustness

A slice computed at $|0\dots0\rangle$ is rechecked, with its operations unchanged, under 100 random computational-basis inputs, 100 random product inputs and, for circuits with rotation angles, 100 random rebindings of every angle. A pair is robust if the slice preserves the criterion in every draw. The gate check takes each of the 21 two-qubit gates of Qiskit's standard library, parameterized gates at a fixed angle, and searches seeded random product inputs, for each direction, for two inputs of one qubit that leave the other in different reduced states, evidence for these gates and not a proof for all.

F. Faults and fault localization

Faults are seeded with ten operators, nine from Muskit [49], QMutPy [50] and García de la Barrera et al. [51], and an angle perturbation of our own. Of the 5,375 mutants, 4 with an empty fault site are excluded, and 2,662 change the measured distribution of at least one qubit. Each such qubit and its mutant form a unit, 5,688 in all, for which we record each slice's size and whether it contains the fault site. Of the 107 entries of Bugs4Q, 11 are circuit-level faults that port mechanically to the current Qiskit release, and seven of them change a qubit's measured distribution, giving 15 units. Running each port records the line that appends each operation, mapped to the original program, so these slices are also reported as lines of the programs their developers wrote.

Mutation-based fault localization ranks the operations of the faulty circuit by how their mutants change its observable behavior, with the Ochiai formula, as Ishimoto et al. do for quantum programs [52], on a sample of 293 units stratified by fault operator and including every Bugs4Q unit. We rank

TABLE III
THE QDG ON EVERY QUESTION OF THE STUDY

Question	Expected	Observed
Sound (RQ1)	every input, Theorem 1	0 wrong slices of 784 pairs, and of 633 on corpus L
Complete (RQ1)	Proposition 1	recall 1.000 of the measured dependences
Size (RQ1)	not minimal	4.8 points above the dependence set, equal to it on 559 pairs
Noise (RQ2)	local noise, Theorem 1	trace distance at most 4.2×10^{-15}
Faults (RQ3)	no guarantee	contains 95.7% of the faults, 17.5 points above a random set of its size
Cost (RQ4)	one pass	at most 7 ms per criterion

TABLE IV
SLICES OF THE 784 CORPUS-A PAIRS FOR RQ1

Rule	Wrong	Dynamic	Recall	Precision	Size
Dependence set[a]	–	–	1.000	1.000	65.6%
QDG	**0**	0	**1.000**	0.923	70.4%
Def-use	216	13	0.718	0.970	51.2%
Classical PDG	204	1	0.741	0.962	53.0%
QPDG [7]	0	0	1.000	0.835	78.6%
QPDG, flow-sensitive[b]	0	0	1.000	0.849	77.5%
Commutation[c]	0	0	0.930	0.967	64.4%
Commutation, symbolic[c]	0	0	0.939	0.966	65.2%
State-based	0	0	0.807	0.972	54.0%

Wrong counts the slices that do not preserve the criterion at $|0\ldots0\rangle$, a lower bound for every rule but the QDG, which is sound for all inputs, and Dynamic those among the 80 pairs in dynamic circuits. Recall and precision are against the dependence set, and size is the mean share of the circuit. [a] The smallest slice that contains every dependence, a reference and not a rule. [b] Exploratory, the variant of Section IV-D. [c] On the 732 pairs where Qiskit's pass applies, 38 of them dynamic.

over the whole circuit and within each slice and report the EXAM score, the share of ranked operations examined before the fault, lower being better.

In the language-model tasks, a model lists the operations that can affect a qubit for 171 items, and names the likely faulty operation of 276 units given the whole circuit or a slice. The models are Claude Sonnet 5.5 and Haiku 5.5 and Qwen2.5-Coder with 32 and 1.5 billion parameters.

G. Cost

RQ4 times the rules on the generated family that grows in depth at eight qubits, on the width sweeps and on corpus L, each circuit in a fresh process limited to 120 s and 4 GiB.

H. Statistics

Five confirmatory hypotheses concern state-based pruning. It is wrong on fewer pairs than def-use (H1), its slices are smaller than QDG slices (H2), at least 95% of pairs keep a correct slice under random product inputs (H3), it loses at most five points of fault containment against the QDG slice (H4), and its slices are smaller where both contain the fault (H5). It starts from the QDG slice and checks every removal at $|0\ldots0\rangle$, so H1 and the directions of H2 and H5 hold by construction, and H3 and H4 are the real tests. We use exact McNemar tests on paired outcomes and Wilcoxon signed-rank tests on paired sizes, with Holm's correction across the five. Effect sizes come with bootstrap 95% intervals and Wilson intervals for proportions, and exploratory intervals resample whole algorithm families. Everything else is descriptive.

VI. RESULTS

Table III gives the answer first. The QDG does what Section IV predicts on every pair the oracle can check and costs milliseconds, but it cannot guarantee to contain a fault, which can delete the very interaction a slice would follow. The subsections answer each question for every rule.

A. RQ1: Which rules give sound and complete slices?

One choice decides soundness, what a controlled gate writes. The classical PDG misses 3,174 dependences over the 784 pairs, and all but 10, 99.7%, follow from reading a control as unchanged. Of the misses, 822 are controlled gates whose control is the criterion qubit, 1,688 reach the criterion only through such a gate, 654 through a control further along the chain, and 10 through classical feedback. Table IV shows that the classical PDG gives a wrong slice for 204 pairs, 26.0% (95% interval 17% to 35% with the 93 algorithm families resampled as clusters). Its control dependence matters only on dynamic circuits, where it is wrong on 1 pair against 13 for def-use.

Many of these wrong slices look right. Of the 204, 88 (43%) give the criterion the whole circuit's outcome distribution in the computational basis, so only 116 pairs, 14.8% of all, are wrong in what a terminal measurement reads. The others are wrong in phase, which a histogram check accepts but which changes the outcome once the slice is composed with later gates or measured in another basis, the limit that phase-flip faults pose to testing [60]. As H1 expects by construction, state-based pruning, which starts from the QDG slice, is wrong on fewer pairs than def-use, 0 against 216 (exact McNemar, Holm $p < 10^{-10}$).

The QDG is sound on every pair and recovers every measured dependence, as Theorem 1 and Proposition 1 predict. The QPDG is sound too, since its slices contain the QDG's, but its entanglement edges add only operations that are not dependences, and its slices grow from 70.4% to 78.6% of the circuit. The flow-sensitive variant still keeps 11.9 points above the dependence set, and a graph with one entanglement edge per two-qubit gate gives exactly the QDG slice on all 779 pairs it applies to, so the excess comes from closing these edges transitively. Under the reading in which a controlled gate writes only its targets, the QPDG is wrong on 10 pairs. On the 633 pairs of corpus L that the oracle can check exactly, the QDG and the QPDG are sound everywhere, while def-use is wrong on 205.

The rules nest. On all 784 pairs the dependence set lies

TABLE V
SLICE SIZE BY CIRCUIT WIDTH (RQ1) AND SLICES THAT STAY CORRECT UNDER OTHER INPUTS (RQ2)

	Mean size, % of circuit					Still correct, %		
Rule	2–3	4–6	7–10	11–14	L	Basis	Prod.	Angles
Dependence set	82.0	75.0	52.7	41.8	–	–	–	–
QDG	83.6	77.7	59.9	51.5	55.8	100.0	100.0	100.0
QPDG	87.6	81.7	71.2	71.0	72.0	–	–	–
Commutation	82.6	75.5	50.1	39.0	43.2	100.0	100.0	97.7
Commutation, symbolic	–	–	–	–	–	100.0	100.0	100.0
State-based	–	–	–	–	–	65.8	26.6	78.1

Widths are in qubits, and L is corpus L, of 15 to 20 qubits, which has no dependence set because measuring one would need more simulation than the oracle can do exactly. Still correct is the share of pairs whose slice, computed at $|0\ldots0\rangle$, preserves the criterion in all 100 draws of random basis inputs, random product inputs and rebound angles, the last on circuits with rotation angles. Commutation is measured on the 732 pairs where Qiskit's pass applies and state-based pruning on the 477 pairs it prunes, and – marks what was not measured.

within the QDG slice and the QDG slice within the QPDG slice, and Table V shows how far apart they are. Restricted to its dependence set, every circuit preserves the criterion at $|0\ldots0\rangle$, so at that input the dependence set is itself a correct slice, the smallest one that contains every dependence. It makes up 65.6% of a circuit on average. The QDG slice keeps only 4.8 percentage points more and equals it on 559 of the 784 pairs, while the QPDG keeps 13.0 points more, 2.7 times as much code that cannot change the criterion. The gap grows with width. At 11 to 14 qubits the QDG slice keeps 51.5% of the circuit and on corpus L 55.8%, while the QPDG stays near 72.0%, because it links the operations of a whole coupling component. Commutation pruning falls below the dependence set, to 39.0% at the same widths, because it is sound for fixed angles but not complete, and can drop an operation where a fault would change the criterion, like the identity gate of line 38 in Fig. 1.

B. RQ2: How much precision beyond the QDG slice is safe?

Both refinements make slices smaller, but only commutation pruning keeps them correct for fixed angles. State-based pruning shrinks the QDG slice by 16.3 percentage points of the circuit (95% CI 15.1 to 17.7) and is smaller on 477 pairs (H2, Wilcoxon, Holm $p < 10^{-10}$). Table V shows the cost. Of the pairs it prunes, the slice stays correct for all random product inputs on only 26.6%, and over all pairs on 55.4% (Wilson CI 51.9% to 58.8%, 41% to 68% with families as clusters), far below the threshold of 95%, so H3 fails.

Commutation pruning is smaller than the QDG slice on 243 of 732 pairs, stays correct on every random basis and product input, and fails only when the angles change, on 4 of 175 pairs of variational circuits. Its symbolic variant, at a mean size of 65.2% against 64.4%, stays correct everywhere, but a recall of 0.930 shows that pruning misses dependences.

Under depolarizing noise at two-qubit error rates of 10^{-3} to $3\cdot10^{-2}$, a pre-registered descriptive analysis, the criterion's states under the QDG slice and the whole circuit differ by a trace distance of at most 4.2×10^{-15} where the simulation is exact, rounding error as Theorem 1 predicts, while the state-based slice differs in 702 of 1,746 exact runs, by up to 0.24, because it removes noisy gates that the program executes [61].

TABLE VI
FAULTS AND SLICES ON THE 5,688 SEEDED-FAULT UNITS, FOR RQ3

Slice	**Contains**	**Size**	**Random**	**Gain**	**EXAM**
Whole circuit	100%	100%	–	–	0.320
QDG	95.7%	76.8%	78.2%	17.5	0.394
QPDG	97.4%	82.9%	84.0%	13.4	0.369
Classical PDG	78.8%	59.0%	60.7%	18.1	0.470
Def-use	77.3%	57.5%	59.2%	18.1	0.471
State-based	82.2%	60.5%	62.4%	19.8	0.440

Contains is the share of units whose slice of the faulty circuit contains the fault site, size the mean share of the circuit, Random the containment of a random set of the slice's size, and Gain the difference in percentage points. EXAM is the mean score of mutation-based fault localization of Section V, lower being better, on the 223 sampled units whose fault every slice contains.

C. RQ3: What does each rule do to fault localization?

Table VI shows that a slice that contains the fault more often is also larger, so containment has to be read against a random set of the slice's size. The QDG slice contains the fault in 95.7% of the units and the QPDG slice in 97.4%, but a random set of the QPDG's size already contains it in 84.0%. Every rule beats a random set of its own size, so containment is not only a matter of size, and per unit of size the QDG does best, 17.5 points above random (family CI 13.2 to 21.8) against 13.4 for the QPDG. All 245 units whose fault the QDG slice misses are omission faults, a removed control (66), a re-targeted operand (91), a removed preparation of an entangling gate (59) or a deleted gate (29). Each deletes the interaction the slice would follow. The QPDG contains 96 of them only because its closure still links operations the faulty program no longer connects.

State-based pruning lowers containment by 13.5 percentage points against the QDG slice (95% CI -14.4 to -12.6, -19.9 to -7.5 with families as clusters), so H4, which allowed a loss of at most five points, fails. In 769 units only the QDG slice contains the fault. A fault that breaks a correlation can make the faulty operation look irrelevant to a single criterion qubit, and a joint criterion over all wrong qubits recovers 691 of these units. Where both slices contain the fault, the state-based slice is smaller by 2.9 operations on average, so H5 holds, but its EXAM score is higher by 0.061.

Slices do not improve mutation-based fault localization. On the 223 units whose fault every slice contains, ranking within any slice reaches the fault after 1.8 to 1.9 operations on average, as ranking over the whole circuit does, and since EXAM divides by the number of operations ranked, the whole circuit scores lowest.

On the 15 real Bugs4Q units whose faulty lines map to the program, the QDG line slices keep every faulty line and 10.3 of 10.7 lines on average, so on these small programs slicing

barely reduces the reading. Def-use, which equals the classical PDG on these unitary programs, loses the faulty line in one of the units.

Language models: Asked to list the operations that can affect a qubit, Claude Sonnet 5.5 and Haiku 5.5 recall 0.84 and 0.84 of the dependences and both Qwen models 0.59, where the QDG slice recalls them all, and Sonnet recalls only 0.61 of those through kickback. Asked to name the faulty operation, no model does better with any slice than with the whole circuit by more than a fraction of a point, and Sonnet has the fault in its top three for 69.9% of the units with the state-based slice, 7.2 points below the QDG slice (95% CI $[-12.3, -2.5]$, exact McNemar $p = 0.006$). A model thus misses dependences that a slicer finds by rule, yet a slice does not help it localize a fault, and a pruned slice hurts it.

D. RQ4: What does it cost?

The QDG slice costs at most 7 ms per criterion at every depth. State-based pruning, which simulates the circuit for every candidate removal, exceeds 120 s for four of the five circuits at the second-largest depth and for all five at the largest, grows exponentially with width, and completes on 39 of the 55 circuits of corpus L. Building the QPDG takes 1.10 s at depth 91 and 21.7 s at depth 380 in the median, because its closure links every pair of operations on a coupling component.

VII. Implications for Comprehension Tools

What a slicer must do: A slicer must let every gate write all its qubits, so a representation that reads the control as unchanged, as the QCPG and order tables built for pattern detection do, should not be read as a dependence graph. A slicer must also follow measured bits, without which def-use is wrong on 13 of the 80 dynamic pairs, and should not add entanglement edges, which add 2.7 times as much code that cannot matter and take seconds to build where the QDG takes milliseconds. Since one loop iteration can put a line in a slice, it should show the iterations that do.

What a developer should not trust: A slice cannot be validated by its outcomes, as 43% of the wrong classical slices and the Bugs4Q example show, so a tool that checks slices should compare reduced states, or at least joint outcomes in more than one basis. Pruned slices answer a narrower question. A commutation-pruned slice is right for one observable at the circuit's angles. A state-based slice is right for one input, stays correct for random inputs on only 55.4% of the pairs, and contains the fault 13.5 points less often. A tool should label either as such and show the QDG slice by default.

What a slice cannot show: Every unit whose fault the QDG slice missed is an omission fault, a removed control, gate or preparation, or a gate moved to another qubit. No slice contains an operation the program lacks, so when a qubit's slice leaves out a qubit the developer expected, that absence is the clue, and listing the qubits each slice contains makes it visible.

What remains open: Slices narrowed where to look but improved neither fault ranking nor a language model's localization, while the models missed dependences that the rule finds. Whether slices shorten the manual reading that developers report does not scale [4] needs a study with developers, which the semantics and measured rules of this paper make possible to design.

VIII. Threats to Validity

Construct validity: The dependence set is measured with two random perturbations per operation at 11 inputs, so it can miss true dependences but never adds false ones, and the QDG's distance to it may be smaller than reported. The dependence set and the wrong slices were checked at $|0 \dots 0\rangle$, which RQ2 extends to random inputs. Each criterion is a single qubit, and Proposition 2 extends the QDG to sets but not the pruning rules. Fault containment measures where a developer would look, not whether a developer finds the fault. The QPDG is reimplemented from prose, with every reading logged before any run and three variants bracketing the plausible ones. Qiskit's pass applies to 732 of the 784 pairs, those without mid-circuit measurements.

Internal validity: Errors in the slicers or the oracle are guarded against by the agreement with Qiskit's simulation, the 24 hand-derived cases, and the nesting of the rules on every pair, which would expose a slicer that drops or adds operations. The soundness proof is mechanized for an abstract model of local operations, and that quantum operations satisfy the model is argued on paper. H1 and H2 were known in advance and hold by construction, and every result outside the five hypotheses is exploratory. The Claude models accept only their default sampling, so their answers can differ between calls.

External validity: The circuits with an exact ground truth have at most 14 qubits in corpus A and 20 in corpus L, and 80 pairs come from dynamic circuits. The proofs hold for every circuit, but the measured distances between the rules may change with width and feedback. Benchmark circuits are not programs written by developers, the 15 Bugs4Q units are the only real faults, and seeded faults may not match real ones in their mix. The gate check uses one angle per gate, the noise analysis depolarizing noise only, and loops that run on the device are not studied.

IX. Conclusion

A quantum slice helps a developer only if it never hides code that can change the qubits asked about. We defined that requirement as preserving the qubits' joint state for every input while keeping every semantic dependence, and showed that the write-all rule meets it. Its slice is the backward light cone extended along classical bits, needs only classical data, classical control and quantum data dependence, and is computed in one pass. A pre-registered study measured what the other rules lose. Reading a control as unchanged gives wrong slices that often reproduce the program's outcomes, entanglement edges add only code that cannot matter, and

pruning trades dependences for size and holds for one observable or input. The QDG slice comes within a few points of the smallest complete slice and costs milliseconds. Two questions remain, whether these slices shorten the manual reading that developers report does not scale, which needs a study with developers, and how to compute them statically on the source developers write, at the scale of real programs and with loops that run on the device.

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